Naive Bayes Letter Recognition

Saxton Calvert
Otto Kanenwisher

Why is this important?

Reading old documents? Ask a computer to do it for you! We need to teach it how though. This is one way to do so!





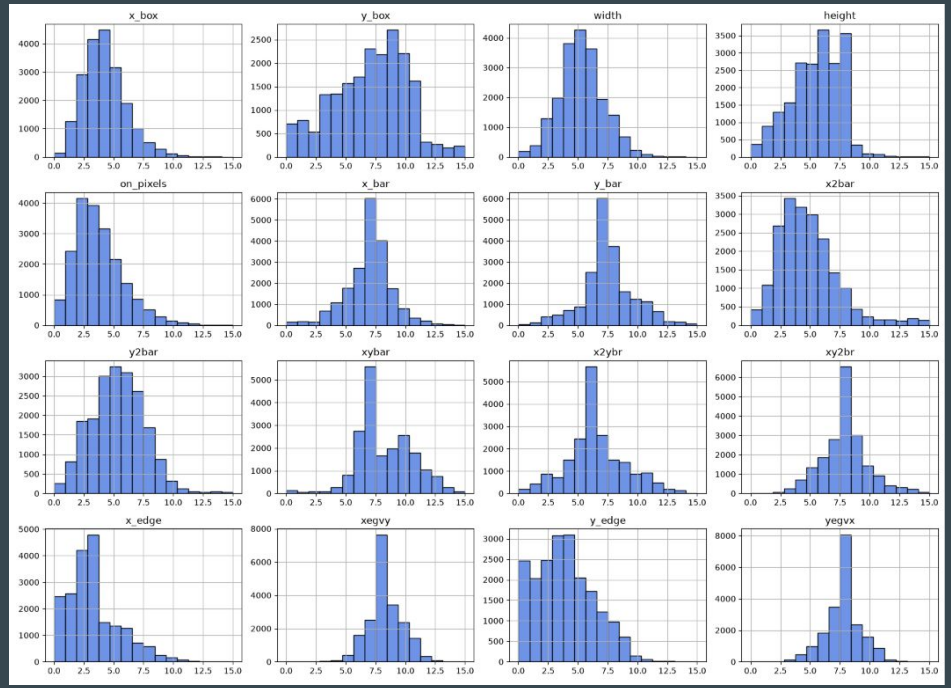
Dataset - Letters as numbers

Because images take up a ton of space, our dataset described letters with numerical features. Other data analysis showed that it worked to classify letters with other models.

	letter	x_box	y_box	width	height	on_pixels	x_bar	y_bar	x2bar	y2bar	xybar	x2ybr	xy2br	x_edge	xegvy	y_edge	yegvx
0	T	2	8	3	5	1	8	13	0	6	6	10	8	0	8	0	8
1	I	5	12	3	7	2	10	5	5	4	13	3	9	2	8	4	10
2	D	4	11	6	8	6	10	6	2	6	10	3	7	3	7	3	9
3	N	7	11	6	6	3	5	9	4	6	4	4	10	6	10	2	8
4	G	2	1	3	1	1	8	6	6	6	6	5	9	1	7	5	10

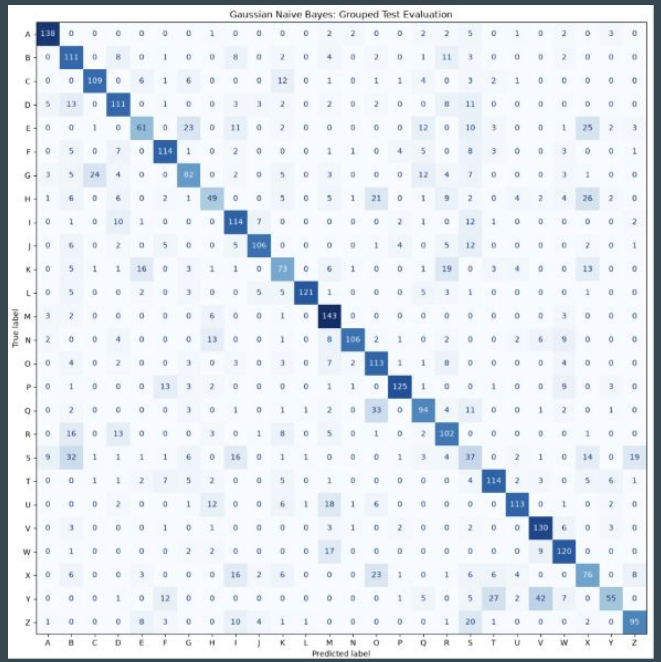
Technique - Gaussian Naive Bayes

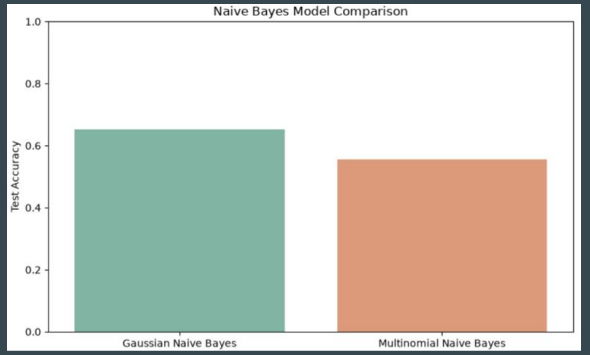
The features aren't counts but they're still numerical, so we anticipated Gaussian Naive Bayes would be the best model. We tested Multinomial just to compare them.



Results - Not very good

Accuracy was only 65.30% with a Gaussian Naive Bayes model, and 55.40% for a Multinomial model. The mixed up letters have similar patterns.

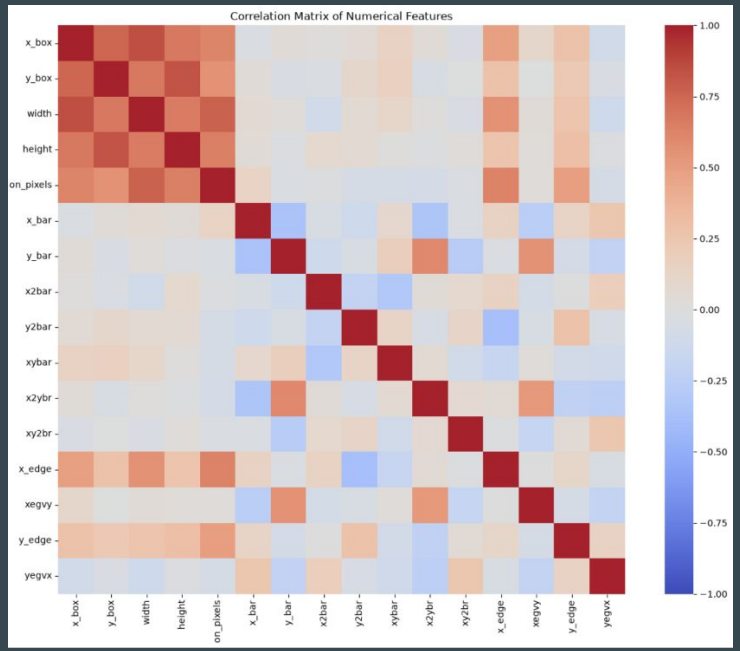





```
Most common grouped-test misclassifications:
Y  V    42
Q  O    33
S  B    32
Y  T    27
H  X    26
E  X    25
G  C    24
E  G    23
X  O    23
H  O    21
```

Results - Why?

Strong feature correlations and the Gaussian model's distributional assumptions may limit performance in addition to shared letter patterns.



Most common grouped-test misclassifications:		
Y	V	42
Q	O	33
S	B	32
Y	T	27
H	X	26
E	X	25
G	C	24
E	G	23
X	O	23
H	O	21